How Far Can General-Purpose LLMs Go in Tigrinya? A Lexicon-Grounded Evaluation with Trivial Baselines

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Abstract

We evaluate eight general-purpose large language models on three lexicon-grounded Tigrinya tasks (part-of-speech tagging, morphosyntactic labelling, and English→Tigrinya translation) using a 5,783-row annotated bilingual lexicon. A fourth task, lexical alignment, is not well enough posed to carry a result, and we release its data and outputs as a resource for reposing it rather than as a benchmark. Our central methodological finding is that such evaluations are easy to over-read: on these tasks a high score is not evidence of Tigrinya competence, because the tasks cannot distinguish competence from much cheaper strategies. A permissive matching rule puts the best model at 85.9% on part-of-speech tagging and 89.4% on morphosyntactic labelling, but scoring the same saved outputs by the first word the model commits to costs up to 26 points, and requiring an exact answer costs up to 89. More decisively, the prompt supplies an English gloss beside the Tigrinya form, and an oracle that sees only that gloss scores 96.5% and 95.9% on the two tasks, which *no model comes within 14 points of*, because only about one headword in twenty is ambiguous given the English. What a model's score on these tasks therefore bounds is what the English already determines, not what the model knows about Tigrinya. On translation all eight models score at or below 0.81 chrF on a 0–100 scale, which is effectively no Tigrinya output at all, while NLLB-200-3.3B, which supports the language, reaches 16.05 on identical items with the same scorer: a factor of twenty. We therefore report a negative result: these tasks, as posed, do not measure Tigrinya competence, and published numbers on them should not be read as doing so. We release our derived evaluation set, the scoring code and the saved model outputs, and argue that the procedure applied here (report every matching rule, bound the task from above as well as below with references that use none of the target language, and quantify prompt echo) requires no resources beyond those an evaluation already has, and should be standard in low-resource evaluation.

1 Introduction

Tigrinya is a Semitic language of Eritrea and northern Ethiopia with approximately 9–10 million speakers and very little machine-readable linguistic infrastructure. This scarcity makes it a natural test of whether general-purpose LLMs generalise beyond high-resource languages, and a difficult one to measure: without established benchmarks it is easy to produce numbers that look like competence.

This paper makes three contributions.

1. **A rebuilt and documented version of an existing lexicon-grounded evaluation set**: 5,783 English-Tigrinya rows (5,022 distinct English headwords, 3,926 Tigrinya forms), each carrying a part-of-speech label, derived from a digitised bilingual dictionary (Union of International Democrats, 2015). The lexicon and the four tasks are not new here; they were introduced by Teklehaymanot et al. (2026), and Section 7 states what differs. What is new is that the set is reconstructible from its source by a released script, with its two digitisation defects corrected and its annotation provenance recorded.

2. **An evaluation of eight LLMs** across POS tagging, morphosyntactic labelling and translation, reported under three matching rules and against references that use no Tigrinya at all. A fourth task in the benchmark, lexical alignment, is reported for completeness but not claimed as a result: 38% of its gold labels are positional fallbacks rather than annotation (Section 5.4).

3. **A negative result about the task design.** The choice of matching rule moves reported accuracy on fixed outputs by up to 26 points

between the two lenient rules and by as much as 89 points once exact match is included, and an oracle restricted to the English gloss scores 96.5% and 95.9% on the two tagging tasks, which no model approaches. The tasks are therefore substantially solvable without the target language, and the numbers they produce, including ours, do not measure Tigrinya competence. We give the procedure that exposes this and argue it should be routine.

This lexicon and these four tasks were introduced by Teklehaymanot et al. (2026), whose retraction has been requested. The results below come from new runs on a rebuilt gold set, not from re-scoring that paper's outputs; Section 7 gives the differences.

2 Related work

2.1 Tigrinya and Ge'ez-script NLP

Tigrinya remains severely underrepresented in NLP: a survey of over 50 studies from 2011 to 2025 finds a field driven almost entirely by discrete resource-creation milestones rather than by modelling advances (Gaim and Park, 2026), a pattern echoed in broader surveys of Ethiopian-language NLP (Tonja et al., 2023). The resources that exist are small and task-specific: domain-specific Tigrinya→English NMT by transfer from other Ge'ez-script languages (Öktem et al., 2020); sentiment transfer from English XLNet, reaching 78.88 F1 on 10k examples (Tela et al., 2020); and two question-answering sets, TiQuAD with 10.6k pairs over 572 paragraphs (Gaim et al., 2023) and TiGQA with 2,685 (Teklehaymanot et al., 2024). Tagging specifically has been approached with neural taggers over the small Nagaoka corpus, where a BiLSTM reaches 92% (Tesfagergish and Kapočiūtė-Dzikienė, 2020); those systems are trained on running text, whereas we evaluate untrained models on dictionary headwords.

Tigrinya appears in some multilingual benchmarks, such as Belebele, a parallel reading-comprehension set in 122 language variants (Bandarkar et al., 2024), and is absent from others: MasakhaPOS, the largest POS dataset for African languages, covers 20 languages under UD guidelines but not Tigrinya (Dione et al., 2023). AfroBench reports the same gap at scale, over 64 African languages and 15 tasks, and finds LLM performance tracking monolingual data availability (Ojo et al., 2025). That is the structural pattern Joshi et al. (2020) documented: representation in NLP venues tracks resource availability. It also bears on how a positive result here should be read, since Blevins and Zettlemoyer (2022) show that English pretraining corpora contain enough incidental non-English text to explain much cross-lingual transfer, so apparent Tigrinya ability need not come from deliberate coverage.

The lexicon and the four tasks used here were introduced by Teklehaymanot et al. (2026), whose retraction has been requested; Section 7 states what this paper takes from that work and how the two differ.

2.2 LLM evaluation for low-resource languages

NLLB-200 added Tigrinya (`tir_Ethi`) to a 200-language translation model (NLLB Team et al., 2022), evaluated on FLORES, whose professionally translated design set the standard for low-resource MT benchmarking (Goyal et al., 2022). Several studies establish that general-purpose LLMs lag purpose-built MT on exactly these languages. Robinson et al. (2023) found NLLB superior to ChatGPT on 169 of 201 FLORES-200 languages (84.1%) by an average of 11.9 chrF, the gap widest for African languages and the Ge'ez script, and report five-shot prompting improving on zero-shot by only 0.88 chrF across 203 directions — close to our own few-shot probe, which moves gemma-7b from chrF 0.16 to 0.66 with exact match unchanged. Hendy et al. (2023) reach the same conclusion for GPT models across eighteen directions, competitive for high-resource and limited for low-resource, and Zhu et al. (2023) find GPT-4 beating NLLB in 40.91% of directions while trailing badly on the long tail. Adebara et al. (2025) extend this to a continent-scale African benchmark and attribute the residual disparities to data inequities rather than model scale. Our Section 5.3 result — three NLLB checkpoints at chrF 14.1–16.1 against eight LLMs at ≤ 0.81 — is this finding at lexicon granularity, where the LLM floor is not merely lower but absolute.

A separate strand asks whether such evaluations are conducted carefully enough. LAraBench benchmarked 33 Arabic tasks across 61 datasets and explicitly computed a random baseline "to determine if the LLMs predictions are not merely the result of chance" (Abdelali et al., 2023). We regard that step as mandatory rather than optional, and Section 5 shows why.

2.3 Trivial baselines and spurious cues

The methodological core of this paper has a direct precedent in natural language inference. Poliak et al. (2018) showed that a hypothesis-only model (one that never sees the premise, a degenerate solution by construction) significantly outperforms a majority-class baseline on a number of NLI datasets; Gururangan et al. (2018) showed the same artefacts let a simple text classifier label the hypothesis alone correctly in about 67% of SNLI and 53% of MultiNLI, concluding that model success "has been overestimated". McCoy et al. (2019) demonstrated that strong test-set scores can rest on fallible syntactic heuristics that collapse on controlled sets, and Niven and Kao (2019) found BERT's 77% on argument reasoning comprehension to be entirely accounted for by spurious statistical cues. Our English-only tagger is a hypothesis-only baseline in this sense: the prompt format `Phrase: {Tigrinya} ({English})` leaves the part of speech largely recoverable from the gloss, and the baseline reaching 60.38 is the lexicon-task analogue of the 67% SNLI result.

For POS tagging specifically, Kann et al. (2020) make the closest argument to ours: weakly supervised taggers reported to perform "almost as well as supervised ones" were evaluated on languages unlike truly low-resource ones, and on 15 genuinely low-resource languages the best model gets fewer than half the words right, with a strongest baseline macro-average of 39.11%. They also note that such methods presuppose "high-coverage and almost error-free dictionaries" precisely the assumption our Section 3.3 annotator notes undermine, where the source dictionary contains mislabelled parts of speech and outright mistranslations. At the level of benchmark design, Reuel et al.'s assessment of 46 best practices across 24 benchmarks identifies stating random performance as a scored criterion, since without it one cannot separate capability from chance or metric design (Reuel et al., 2024).

2.4 Scoring rules, answer extraction and prompt artefacts

Our finding that the matching rule moves reported accuracy on fixed outputs by up to 26 points between lenient rules, and by as much as 89 points once exact match is included, belongs to a growing literature on evaluation mechanics. Tam et al. (2024) show that constraining LLMs to structured output formats significantly degrades measured reasoning, and disentangle format errors from content by using an LLM as a "perfect parser" rather than a regex the mirror image of our problem, where permissive parsing manufactures competence instead of destroying it. Kirouane and Petrocheilos (2026) document the same failure mode concretely in a low-resource setting: a scorer that scanned the whole response instead of the requested answer line produced a $+29.8$ pp artefact, one-directional against verbose outputs, and they conclude that "any benchmark that requests an answer format must score that format first and report how often it was absent". Our first-token rule is that recommendation applied; our exact / first-token / set-intersection triple reports the spread it would otherwise hide.

The prompt-echo column in Section 5.2 relates to known label-copying behaviour. Zhao et al. (2021) show that few-shot LLM accuracy is destabilised by bias towards answers appearing in the prompt or common in pretraining, varying "from near chance to near state-of-the-art" with prompt format alone. When the gold distribution is 53.7% `noun` and the prompt example contains the word `noun`, a model that copies the example is scored correct by construction under set-intersection gemma-7b does this on 97.4% of answers. We therefore treat echo rate as a reportable diagnostic rather than a footnote.

2.5 Metrics, agreement and dictionary sources

We report chrF (Popović, 2015) because character n-gram F-score is tokenisation-independent and so appropriate for Ge'ez-script targets averaging 1.18 whitespace tokens, where word-level BLEU (Papineni et al., 2002) is near-undefined; following Post (2018), who argues that BLEU is a parameterised family whose configurations are not comparable across papers unless the scheme is pinned down, we report character-level BLEU alongside it. Agreement is Cohen's κ (Cohen, 1960), reported beside raw observed agreement and the filled-label-only variant rather than as a single coefficient, because κ is hard to interpret under class imbalance — the "kappa paradox" — and absolute thresholds are not meaningful without a contextual reference (Wong et al., 2021).

Digitised bilingual dictionaries are a standard bottom-up resource for languages lacking parallel corpora: Wickramasinghe and de Silva (2023) argue that for low-resource pairs it is "more feasible to move in the bottom-up direction where finer granular pairs such as dictionary datasets are developed first", while Ve'rdd addresses the complementary

problem of grassroots paper dictionaries edited by many hands, which need re-evaluation rather than direct ingestion (Alnajjar et al., 2020). Our source shows exactly the defects that motivate that work — a direction-swapped second block, inline gender markers inside the target field, and annotator-flagged mistranslations — which Sections 3.1 to 3.3 document rather than silently clean.

3 Dataset

3.1 Source and construction

The lexicon was digitised from the Swansea Tigrinya–English dictionary (Union of International Democrats, 2015), with additional entries contributed by native-speaker linguists, and carries the source dictionary's own part-of-speech abbreviations. The digitised source holds 7,234 rows concatenated from two directions, English→Tigrinya followed by Tigrinya→English, under a single pair of column headers; in the second block the column named `English` therefore holds Ge'ez script. We detect that block by script, reorient it, and merge both directions, keeping distinct senses of one form as separate items — *estimate* → ግምት is retained as both noun and verb, which yields **5,783 gold rows** over 5,022 distinct English headwords and 3,926 distinct Tigrinya forms. Counts of headwords are sensitive to the definition: splitting the comma-separated variants that some entries carry in the English field gives 4,327 single-word glosses instead. We count whole field values throughout, which is what the baselines and the ceiling in Section 4.2 key on. Three evaluation set sizes follow from this and are not interchangeable: 5,783 gold rows, **5,775 items** for translation and lexical alignment after each task drops duplicate pairs and rows with an empty side, and **5,701 items** for the two tagging tasks, which were run against a further reduction representing each multi-sense entry by its first sense. Appendix C gives the row-level derivation, the provenance of the source file, and the reconstruction of the 5,701-item set from the released gold.

3.2 What the annotations do and do not contain

The POS labels are the source dictionary's, not independently re-annotated by us. They are coarse: `noun` accounts for 3,109 of 5,783 labelled entries (53.8%), `verb` 1,542, `adjective` 879.

Morphosyntactic features are **sparse**. Only 183 entries carry a morphosyntactic label that differs from the POS label at all; the remaining 5,600 (96.8%) are identical to it. The 183 are sparser still than that suggests: 88 carry gender, number or plurality (38 `noun, masculine`, 35 `noun, feminine`, 13 `noun, plural`, 2 `uncategorized`), and the other 95 are `adjective, noun-derived`, a derivational note rather than an inflectional feature. We therefore do not treat morphosyntactic labelling as a distinct linguistic dimension, and report it in Section 5.2 as what it measures: a second POS task with a different prompt.

3.3 Inter-annotator agreement

A 500-item subset of the lexicon was annotated independently by two of the authors for part of speech, gender, number, category agreement and lexical alignment. Both are native speakers of Tigrinya. Each assigned labels separately using the annotation guide, and the two sheets were then compared to locate disagreements. The final sheet was compiled from those two sheets: **no third annotator took part, and no party outside the author group adjudicated**. We state this because it bounds what the coefficients below can mean, and because the compiled sheet is sometimes read as an independent adjudication, which it is not.

The records are released as `POS_english_to_tigrigna_Annotated.xlsx` the two annotation sheets, the compiled sheet and the annotation guide that defines each value and `scripts/compute_iaa.py` recomputes every figure below from them.

The workbook carries no edit history: the annotators exchanged their sheets as messaging attachments that have since expired, so the labels are intact but the order and timing in which they were entered are not recoverable.

The five dimensions are not of one kind. The guide's instruction is to "verify and annotate": gender, number, category agreement and lexical alignment are assigned from scratch, while the part-of-speech label arrives pre-filled from the source dictionary and is verified. The two annotators used that latitude very differently — Annotator 1 corrected the dictionary's part of speech on 25 of 500 items, Annotator 2 on none — so all 25 part-of-speech disagreements are of one shape: one annotator exercising the verification the guide asks for, the other accepting the dictionary.

An annotator sometimes left a cell empty where

Dimension	κ	Observed
Part of speech	0.9076	0.9500
Gender	0.8079	0.9240
Number	0.6762	0.9280
Category agreement	0.5912	0.9000
Lexical alignment	0.8487	0.9480

Table 1: Inter-annotator agreement on the 500-item subset, blanks counted as disagreements. All five dimensions are computed over all 500 items.

the other assigned a label. Those are not matching judgements, so we count them as disagreements and report Cohen's κ over all 500 items, with a blank treated as its own category (Table 1).

Restricting instead to the items both annotators labelled raises every dimension — 0.9076, 1.0000, 1.0000, 0.9815 and 0.9850 on 500, 420, 428, 445 and 474 items. We report the stricter figures because the difference between the two columns is largely a difference in what each annotator left blank: Annotator 1 omits 32 gender, 32 number and 46 agreement labels that Annotator 2 supplies, against 6, 4 and 3 in the other direction. Discarding those rows would hide the main source of divergence.

Three properties bound what these coefficients support, and Appendix A gives the evidence for each. The compiled sheet is a selection between the two existing columns, not an independent third judgement, so it cannot serve as an adjudicated gold standard. Agreement on filled labels is near-total: where both annotators assigned a gender or number value they never differ, so the divergence is in coverage rather than in choice of label, which is why κ falls once blanks count. And the subset is not a random sample but the alphabetical head of each category: 61% adjective against a corpus that is 54% noun and 25% verb, and **drawn from no verb entries at all**. The coefficients therefore describe this subset and are not an estimate for the corpus; establishing one needs a stratified re-annotation covering verbs, by annotators outside the author group, which we leave to future work (Section 8).

The workbook also carries the annotators' free-text notes on 201 of the 500 items, documenting mislabelled parts of speech in the source dictionary, inflected forms given as citation forms, and outright mistranslations (*at least* → ብብዚሑ, which means *at most*). These support the caveat in Section 3.2 that the POS labels are the source dictionary's and were not independently verified.

4 Experimental setup

Eight models, all loaded through the Hugging Face `transformers` interface (Wolf et al., 2020) and run on a single compute node with four A100 GPUs, 32 CPU cores and 128 GB of RAM (Table 2).

A later run on newer models, and why it is not in the tables. The eight models of Table 2 are those the evaluation was run on in full. They are dated: none exceeds 7B parameters and three cannot follow an instruction. We later began a run on nine newer and larger checkpoints, which reached the time limit after completing part-of-speech tagging only; its log and its saved generations are released with the rest. We do not put it in the tables because it covers one of four tasks, but it does not sit quietly in the logs either, because its best model scores above ours: Falcon3-10B-Instruct reaches 82.62 under the task script's own scorer, against mistral-7b's 78.76 on the same rule.

That comparison illustrates this paper's argument rather than undercutting it. Falcon3-10B emits its chat template into every one of the 5,701 answers (`<|assistant|> adverb`), so on the first-token rule this paper reports it scores **0.00**; strip the prefix and the same outputs score 80.25; score them by set intersection and they reach 82.76. One model, one set of generations, and a range of 82 points depending on how the text is handled. A larger model does not escape the measurement problem, and the ceiling of Section 4.2 applies to it unchanged: 82.76 is still below the 96.49 an English-only oracle reaches, so the higher score is not evidence that it read the Tigrinya.

We do not compare architectures. All five causal models are instruction-tuned and all three sequence-to-sequence models are raw pretrained checkpoints; the two factors are perfectly confounded in this model set and no result here can separate them. The three base seq2seq models receive span-infilling prompts matched to their pretraining objective; they nonetheless produce almost no usable output, which we report as a property of those checkpoints on this task rather than of their architecture.

Generation sets `max_new_tokens=128` and `min_new_tokens=1` and nothing else, so each checkpoint decodes under its own `generation_config.json`. We state this

Label	Checkpoint	Arch.	Inst.
gemma-2b	`google/gemma-2b-it` (Gemma Team, Google DeepMind, 2024)	causal	yes
gemma-7b	`google/gemma-7b-it` (Gemma Team, Google DeepMind, 2024)	causal	yes
mistral-7b	`Mistral-7B-Instruct-v0.2` (Jiang et al., 2023)	causal	yes
qwen-7b	`Qwen1.5-7B-Chat` (Bai et al., 2023)	causal	yes
falcon-7b	`falcon-7b-instruct` (Almazrouei et al., 2023)	causal	yes
mt5-small	`google/mt5-small` (Xue et al., 2021)	seq2seq	no
mt5-large	`google/mt5-large` (Xue et al., 2021)	seq2seq	no
byt5	`google/byt5-small` (Xue et al., 2022)	seq2seq	no

Table 2: Evaluated models. Architecture and instruction-tuning status are perfectly confounded in this set.

precisely because it is not uniform across the roster: Qwen1.5-7B-Chat ships `do_sample=true` with temperature 0.7 and `top_p` 0.8, so qwen-7b was sampled rather than decoded greedily, and its numbers carry sampling variance that the others may not. We did not apply each instruction-tuned model's chat template; prompts were passed as plain text, which is one reason the verbose and off-format answers of Section 4.1 are so common. Both choices are limitations rather than design decisions, and a rerun that fixed decoding to greedy and applied each chat template would be a cleaner measurement of the models; neither affects the paper's claims, which concern what the tasks can measure on fixed outputs. `xlm-roberta-base` (Conneau et al., 2020) was included initially but fails on all tasks (its prompts contain no `<mask>` token) and is excluded.

4.1 Scoring

Model answers are frequently verbose: gemma-7b's median POS answer is 14 words. We report three rules.

- **exact**: the cleaned answer equals the gold label.
- **first-token**: the model's first word is a gold label word. This is the answer the model commits to, and is our headline rule.
- **set-intersection**: any word of the answer is a gold label word. This is permissive: an answer naming several parts of speech is correct as soon as one fits. We report it only to show the gap.

4.2 Baselines

The prompt supplies the English gloss alongside the Tigrinya form (`Phrase: {Tigrinya} ({English})`), so a system can score well without using the Tigrinya at all. We therefore bound the two tagging tasks from below and from above with references that see no Tigrinya whatsoever.

- **Majority**: always answer the most frequent gold label (`noun`).
- **English-only tagger**: an NLTK (Loper and Bird, 2002) English POS tagger applied to the gloss alone. This is what one off-the-shelf English system achieves, and is a lower bound on what the gloss makes available.
- **English-only ceiling**: an oracle that sees only the English string and answers with the most frequent gold label that headword carries anywhere in the evaluation set. This is the best any English-only system can reach, and we report it because the lower bound alone understates how much of the task the gloss already determines.

The ceiling is high: 96.49 on part-of-speech tagging and 95.86 on morphosyntactic labelling. Only 189 of 5,025 headwords (3.8%) carry more than one gold part-of-speech label, and 212 (4.2%) more than one morphosyntactic label. Those headwords account for 439 of the 5,701 items (7.7%) on part of speech and 499 (8.8%) on morphosyntax, and they are the only items on which the Tigrinya form can supply information the gloss does not already determine. A model scoring below the ceiling has not shown that it used the Tigrinya.

Differences against the tagger baseline are tested with a one-sided McNemar test at $n = 5{,}701$. The test must be paired: the baseline is a system scored on the same items as the model, so the two sets of outcomes are correlated and only the discordant items carry information. Treating the baseline rate as a known constant and applying a binomial test, as is sometimes done, understates the variance of the difference and so overstates significance; on our data it moves the smallest margin by an order of magnitude ($p = 0.003$

Model	exact	**first-tok**	set-int
mistral-7b	33.71	**81.56**	85.91
gemma-7b	0.00	**79.39**	82.44
gemma-2b	71.71	**72.86**	73.29
falcon-7b	26.94	**40.64**	44.45
qwen-7b	17.00	**17.00**	17.07
mt5-large	0.00	0.09	1.75
byt5	0.00	0.00	0.05
mt5-small	0.00	0.00	0.00
Majority (`noun`)		**53.73**	
English-only tagger		**60.38**	
English-only ceiling		**96.49**	

Table 3: POS tagging accuracy (%) under three matching rules, against the two English-only references. No model reaches the ceiling.

Model	**first-tok**	set-int	echo
gemma-2b	**64.59**	78.02	1.2%
gemma-7b	**63.32**	89.42	**97.4%**
mistral-7b	**54.92**	81.09	3.7%
falcon-7b	**39.17**	76.28	19.5%
qwen-7b	**34.33**	54.45	0.4%
mt5-small	0.86	2.49	0.0%
mt5-large	0.11	0.74	0.0%
byt5	0.00	38.55	69.7%
Majority	**52.38**		
English-only tagger	**61.57**		
English-only ceiling	**95.86**		

Table 4: Morphosyntactic labelling accuracy (%). The final column is how often an answer contains the prompt's own example. No model reaches the ceiling.

against $p = 0.032$ for gemma-7b on morphosyntax). `scripts/task_ceiling.py` computes the ceiling, the ambiguity counts and both tests.

5 Results

Three tasks carry results; Section 5.4 reports the fourth, whose gold labels are too unreliable to support a conclusion.

5.1 POS tagging ($n = 5{,}701$)

Three models exceed the tagger baseline, each by a margin that is unambiguous ($p < 10^{-50}$): mistral-7b by 21.19 points, gemma-7b by 19.01 and gemma-2b by 12.49. Falcon-7b and qwen-7b fall below the majority baseline; all three seq2seq checkpoints are at or near zero.

No model reaches the English-only ceiling. The best, mistral-7b, falls 14.93 points short of it; gemma-7b 17.10 and gemma-2b 23.63. Since the ceiling is what a system achieves from the gloss alone, the ordering above is not evidence that any model used the Tigrinya form. It is consistent with every model using the gloss and using it worse than an English-only oracle would.

The exact column shows how brittle these numbers are. Gemma-7b scores 0.00% exact and 79.39% first-token: it always answers in a sentence, never a bare label. A reader given only one of these columns would draw opposite conclusions.

5.2 Morphosyntactic labelling ($n = 5{,}701$)

Because the gold is 96.8% identical to the POS label (Section 3.2), this is effectively a second POS task.

Only two models exceed the tagger baseline of 61.57: gemma-2b by 3.02 points ($p = 1.6 \times 10^{-4}$) and gemma-7b by 1.75 ($p = 0.032$). Both margins reach significance at $n = 5{,}701$, and both are small; gemma-7b's is marginal, and would not survive a correction for testing eight models. The remaining six fall below the baseline, and five below the majority baseline of 52.38.

Against the ceiling the picture is uniform: **the best model is 31.28 points below what the English gloss alone makes available**, and the eight models span 31 to 96 points below it. Whatever separates gemma-2b from mistral-7b here, it is not evidence of morphosyntactic analysis of Tigrinya.

The final column measures how often an answer contains the prompt's own example, `preposition, noun, singular`. Gemma-7b echoes it in 97.4% of answers; because the example contains `noun`, and `noun` is the majority gold label, set-intersection scores those echoes as correct. Its apparent 89.42 is largely an artefact of copying the prompt. ByT5's 38.55 is the same artefact at 69.7% echo, and drops to 0.00 under first-token.

5.3 Translation, English→Tigrinya ($n = 5{,}775$)

Tigrinya references average 1.18 whitespace tokens, so word-level BLEU-4 is near-undefined; we report chrF and character-level BLEU (Table 5).

All eight general-purpose models are at the floor. NLLB-200, for which Tigrinya (`tir_Ethi`) is a supported target language, reaches chrF 16.05 on

Model	chrF	$BLEU_c$	$BLEU_w$	exact
NLLB-3.3B	**16.05**	**14.23**	0.24	**16.24**
NLLB-1.3B	15.01	13.29	0.23	15.00
NLLB-600M	14.11	12.59	0.25	14.42
byt5	0.81	0.01	0.00	0.05
gemma-2b	0.39	0.01	0.01	0.07
mistral-7b	0.26	0.00	0.00	0.03
mt5-small	0.20	0.04	0.02	0.05
gemma-7b	0.16	0.00	0.00	0.03
mt5-large	0.13	0.04	0.01	0.02
falcon-7b	0.13	0.00	0.00	0.03
qwen-7b	0.04	0.00	0.00	0.00

Table 5: Translation quality. All values on a 0–100 scale.

identical items with an identical scorer, roughly twenty times the best LLM.

The comparison is not like-for-like, and the differences all favour NLLB. It is a dedicated translation model given a source sentence and a target language code, decoded with five-beam search; the LLMs are prompted in English, decode under their own defaults (Section 4.1), and have their answers scored leniently, any word of a verbose reply counting as a hit against any gold synonym. We draw only the conclusion this supports: a purpose-built model that covers the language produces recognisable Tigrinya on these items and the general-purpose models produce almost none, a gap of a factor of twenty that no plausible decoding or prompting advantage would close. It does not license a ranking of NLLB against these LLMs as systems.

Note that word-level BLEU-4 is 0.24 for NLLB-3.3B, which translates 16.24% of items exactly. **Word-level BLEU is not usable at lexicon scale**, and a paper reporting it would show a working system as scoring zero.

Our own scorer supplied a further instance of the effect this paper reports. Many gold entries end in a trailing comma, and splitting them on commas without discarding the empty final field admitted the empty string as a valid answer, so a model that generated nothing scored an exact match. This gave qwen-7b an exact-match rate of 0.28, the highest of the eight models and all 16 of its apparent successes empty generations. The released scorer drops empty candidates and the table reports 0.00; no other model was affected and no chrF or BLEU value changes. We report the defect rather than silently correcting it because a one-line omission in a scorer, not any property of a model, produced the best exact-match score in the table.

We verified the LLM failure is not one of output format. On a 300-item random sample (seed 0, exemplar phrases excluded), with gemma-7b alone, few-shot prompting with eight exemplars and an explicit format constraint moved chrF from 0.16 to 0.66, and exact match stayed at 0.00 on every arm of that probe; extracting the Ge'ez span from verbose answers recovered the reference in 2 of the 5,775 full-set items. (Gemma-7b's 0.03 in Table 5 is the full set, where 2 items match; the probe's 0.00 is the 300-item sample, where none does.) The median character-level similarity between model output and reference is 0.000: the models emit Ge'ez-shaped strings that are not the target words (ባህል for ብዙሕ, ልልል as repetition).

5.4 Lexical alignment

We report this task for completeness but do not draw conclusions from it: 2,208 of 5,783 alignment labels (38%) are positional fallbacks (`AlignmentSource: auto`) rather than annotation. Format accuracy ranges from 0.0383 (mt5-large) to 0.5484 (gemma-2b); semantic accuracy from 0.0012 (mt5-small) to 0.8876 (mistral-7b). The two measures diverge widely for individual models, which makes the task ill-posed as currently scored.

We release its data and outputs nonetheless: the 3,575 manually verified rows are marked as such, so restricting to them or annotating the remaining 2,208 would give a task this one cannot, and the divergence is itself another instance of the paper's point, two defensible scorers disagreeing on fixed outputs by more than the models differ.

6 Discussion

The matching rule is a reporting decision, not an implementation detail. The swings in Section 5.1 are not a property of the weaker models: the largest belong to the models that answer in fluent sentences, because a sentence contains the gold label without equalling it. Scoring the same saved outputs three ways costs gemma-7b everything it scored. A paper that reports one number without naming the rule has not reported a result, and reporting the triple costs nothing once the generations are saved.

Bound the task from above, not only from below. A lower-bound baseline tells you whether a model beats a weak system; it does not tell you whether the task needs the target language. Three models clear both lower baselines on POS, which reads as cross-lingual competence until the oracle is added and the best of them sits 14.93 points beneath it (31.28 on morphosyntax). Reporting only the lower bounds would have let this paper claim a result it does not have. The oracle costs one pass over the gold labels, and we recommend it wherever a prompt carries a cue in a higher-resource language.

Prompt echo is a measurable confound. Gemma-7b reproduces the prompt's example in 97.4% of its morphosyntax answers, and because the example contains the majority gold label, a permissive scorer rewards it. Echo rate belongs in any report whose prompt contains a worked example.

What we can claim about Tigrinya competence. On the two tagging tasks, nothing: the tasks cannot separate a model that reads Tigrinya from one that reads the English gloss and answers worse than an oracle would. Translation is the one task here not solvable from the prompt's English, and there the result is unambiguous — no model translates into Tigrinya, while NLLB-200, which supports the language, is more than an order of magnitude better on chrF. For low-resource languages, model selection dominates model size; and even NLLB reaches only 16.24% exact on dictionary headwords, so the difficulty is real rather than an artefact of our setup.

7 Relation to prior work, and reproducibility

This lexicon and these four tasks were introduced by Teklehaymanot et al. (2026), whose retraction has been requested.

The results reported here are not a re-scoring of that paper's outputs. They come from new runs carried out in September 2026, against a gold set rebuilt from the source lexicon, and the two differ in their inputs as well as in their scoring. The earlier runs evaluated 3,899 items per tagging task; these evaluate 5,701. The earlier gold carried the source dictionary's parenthetical notation, so an expected tag was stored as `(adv) adverb` rather than `adverb`, and an entry was presented with all of its comma-separated Tigrinya senses rather than one. Prompts built from those entries differ, and so do the generations they elicit. No number is carried over, and the two sets of results are not alternative measurements of the same quantity.

The lexicon, the gold label files, the saved model outputs, the annotation workbook and the scoring code accompany this submission as supplementary material, and will be released publicly on acceptance. Every figure in this paper is recomputed from those artifacts by a released command, and Appendix B lists each table, the command that regenerates it and the artifact it reads. The supplementary material also holds the raw generation for every item, stored per task and per model, so a scoring rule other than the three reported here can be applied to the same outputs without re-running a model.

8 Limitations

- POS labels are the source dictionary's, verified but not independently re-annotated. The agreement coefficients in Section 3.3 describe a 500-item subset that is alphabetical and drawn from no verb entries, annotated by two of the authors rather than by annotators external to the work, and compiled without a third adjudicator. They are not an estimate for the corpus, and a stratified re-annotation covering verbs, by annotators outside the author group, would be needed for that.

- No edit history survives for the annotation. The annotators exchanged their sheets as messaging attachments, and those attachments have since expired, so the files that would carry a version history are no longer retrievable. The labels are intact in the released workbook; what cannot be evidenced is the order and timing in which they were entered.

- Morphosyntactic labels differ from the POS label on 183 of 5,783 gold rows, of which 88 carry gender, number or plurality.

- Lexical alignment gold is 38% positional fallback; Section 5.4 is not a result.

- Entries are dictionary headwords, not running text; performance here does not predict sentence-level performance.

- The model set cannot separate architecture from instruction-tuning. An instruction-tuned

seq2seq model (`flan-t5`, `mt0`) would be needed. It is also dated: none of the eight exceeds 7B parameters, and three are span-denoising checkpoints that cannot follow an instruction at all. A later partial run on newer and larger models (Section 4.1) scores above our best model on POS, which we report rather than leave in the logs, though the ceiling result means a higher score on this task is not evidence of Tigrinya competence.

- Decoding was each checkpoint's own default rather than a uniform greedy setting, so qwen-7b was sampled at temperature 0.7, and no chat template was applied to the instruction-tuned models. Single run, no seed averaging.

- Both tagging tasks are substantially solvable without Tigrinya: an English-only oracle reaches 96.49 and 95.86, and under 5% of headwords are ambiguous given the gloss. This is a property of the task design, not of the models, and it bounds what any result on these tasks can mean. A task that withheld the gloss, or that drew its items from the ambiguous subset, would measure something the English does not already determine; we have not run that experiment.

- The ceiling is computed from the evaluation set itself, so it is an oracle rather than a system: it knows the label distribution of each headword. It bounds what the English makes available on these items, not what an English-only system would achieve on unseen data.

Ethics statement

The annotators are two of this paper's authors, both native speakers of Tigrinya, who annotated as part of their own research and received no separate compensation; no external or crowd workers were engaged, and no personal data was collected. The source material is a published bilingual dictionary and the evaluation items are dictionary headwords, which carry no information about individuals.

Each model was run once, with no seed averaging, under the decoding settings of Section 4.1. Evaluation used one node with four NVIDIA A100 80GB GPUs. The reported results come from two jobs: 66.3 hours wall clock for part-of-speech tagging, morphosyntax and lexical alignment over the eight models, and 13.9 hours for the translation rerun, about 321 GPU-hours together. The partial run on the newer roster (Section 4.1) consumed a further 288 GPU-hours before reaching its time limit. Every number in this paper is then recomputed from the saved outputs on CPU in under a minute, which is the point of releasing the generations.

chrF and character BLEU are computed with NLTK rather than sacreBLEU, so our chrF values are not directly comparable to sacreBLEU figures reported elsewhere; every comparison we draw is within this paper, between outputs scored by the same implementation, including the NLLB comparison. Package versions are listed in the released requirements file.

A risk specific to this work: a benchmark that reports high accuracy on a low-resource language can be cited as evidence that the language is served when it is not, which may divert effort from building the resources it actually needs. That is the failure this paper documents in its own earlier numbers, and it is why we report the task ceiling alongside every score.

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A Annotation records in detail

What the released workbook does not carry is the history of how it was filled. The annotators exchanged their sheets as messaging attachments, and those attachments have since expired, so the files that would carry an edit history are no longer retrievable. The labels themselves are intact and are what the workbook contains; what is missing is the order and timing in which they were entered. We note this because the guide was applied unevenly in ways the figures below expose, and a record of the process would have shown whether that reflects a change of convention partway through.

The five dimensions are not of one kind, and the guide says so. Its instruction is to "verify and annotate" the dictionary entries: gender, number, category agreement and lexical alignment are assigned from scratch against the definitions the guide gives, while the part-of-speech label arrives prefilled from the source dictionary and is verified. The guide treats it as given material elsewhere too, defining category agreement as whether the Tigrinya entry matches "the English item and its POS label".

The two annotators used that latitude very differently. Annotator 1 corrected the dictionary's part of speech on 25 of 500 items; Annotator 2 corrected it on none, letting the source label stand everywhere. All 25 part-of-speech disagreements are therefore of one shape: one annotator exercising the verification the guide asks for, and the other accepting the dictionary.

Three properties of this subset bound what the coefficients support.

The compiled sheet is not an independent third judgement. This follows from how it was produced, and the records bear it out. The two annotators differ on 28 cells in total: 25 part of speech, one category agreement and two lexical alignment. On the 25 part-of-speech items the compiled sheet reproduces Annotator 2 on all 25 and Annotator 1 on none; on the other three it reproduces Annotator 1. **On none of the 28 does it record a label that neither annotator assigned.** Since Annotator 2's part-of-speech column is the source dictionary's throughout, the compilation rejected all 25 of Annotator 1's proposed corrections and deferred to the dictionary in every case. It selects between the two existing columns rather than adding a judgement of its own, so agreement with it is not evidence about either annotator, and it cannot serve as an adjudicated gold standard.

A third annotator would not repair this. The coefficients in Table 1 are computed between the two annotation columns and would be unchanged by a third. The divergence between those columns is in any case not a matter of conflicting labels: on gender and number the annotators never differ on an item both labelled, and what separates them is how often each left a cell empty. What the subset would need is a fresh annotation, stratified over the categories and including verbs, by annotators outside the author group; we report that as future work in Section 8 rather than adjudicating the present records after the fact.

Agreement on filled labels is near-total. Where both annotators assigned a gender or number value, they never differ — 0 disagreements across 420 and 428 items. The divergence between them is entirely in coverage, not in choice of label, which is why κ falls as far as it does once blanks count.

The subset is not a random sample. Tracing each item to its source sheet shows the 500 are the alphabetical head of each category: all 9 prepositions, all 10 pronouns and all 6 interjections, then 305 adjectives, 150 nouns and 20 adverbs from the tops of those sheets. The corpus is 54% noun and 25% verb; this subset is 61% adjective and **is drawn from no verb entries at all**. The only part-of-speech cells reading `Verb` anywhere in the records are seven of Annotator 1's corrections to noun headwords. The coefficients therefore describe agreement on this subset and are not an estimate for the corpus. Establishing one requires a stratified re-annotation covering verbs, which we leave to future work (Section 8).

B Reproducing each table

The supplementary material also contains the raw generation for every item, stored per task and per model, so any scoring rule other than the three reported here can be applied to the same outputs without re-running the models.

`$ROSTER` denotes the eight evaluated models, which the scoring script takes as an explicit argument so that a reported table is reproduced over exactly that set; the task scripts default to the same roster, so a full re-run reproduces the published labels rather than the newer ones.

C Dataset derivation

This appendix records the row-level derivation summarised in Section 3.1. Every count is reproducible from the released code: `build_merged_eval_set.py --dry-run --verify` rebuilds the gold files from the source CSV and compares them against what is released, and `build_sense_split_eval_set.py --verify` does the same for the 5,701-item set.

Source and provenance. The digitised file `Combined_POS_Lexicon.csv` holds 7,234 rows: 3,587 English→Tigrinya entries followed by 3,647 Tigrinya→English entries. It is a flattening of the working annotation workbook, whose eleven part-of-speech sheets hold the same 7,234 rows with the same per-category counts (3,929 noun, 1,784 verb, 993 adjective, 194 adverb, and so on). The flattening lifts the inline gender and number markers that the workbook keeps inside the Tigrinya field `(nf)` ወካሲት, ከሳሲት out into the label column, which is the origin of the finer categories `(nm) noun masculine`, `(nf) noun feminine` and `(npl) noun plural`, which 107 export rows carry. Compared as multisets of (English, Tigrinya) pairs, the two files differ on 247 rows; stripping the parenthesised markers from both reconciles all but 12 of those, so the markers account for the difference without the 247 being the marked rows themselves. Twenty-four rows of the gold files still contain an inline marker, including translation references such as `(af)` ኣደናቒት: these are residual and should be stripped, and we note them rather than correct them silently so that the released gold matches the outputs the models were scored on. The direction swap is present in the workbook as well, in 3,635 of its 7,234 rows (50.2%), so it originates in digitisation rather than in the export.

7,234 counts rows, not independent annotations. Of the 3,646 unique reverse-direction pairs, 1,448 (39.7%) also occur in the forward block.

From 7,234 to 5,783. Merging the two directions deduplicates on the (English, Tigrinya, features) triple rather than on the pair alone, so that distinct senses of one form survive: *estimate* → ግምት is kept as both noun and verb, and *grocer* → በዓል ድኳን as both `noun` and `noun, masculine`. Expanding comma-separated senses gives 5,864 atomic pairs.

From 5,783 to 5,775. Each task script applies its own pair-level deduplication when loading the gold,

Reported in	Command (in `scripts/`)	Artifact read
§3.1 counts	`build_merged_eval_set.py --dry-run --verify`	`Combined_POS_Lexicon.csv`
§3.1 5,701 set	`build_sense_split_eval_set.py --verify`	`*_sense_split.json`
§3.3 agreement	`compute_iaa.py`	`*_Annotated.xlsx`
§4.2 ceiling	`task_ceiling.py`	`*_sense_split.json`
§5.1, §5.2	`rescore_tasks.py --models $ROSTER`	`rescored_2026-09-28.json`
§5.3 LLMs	`compute_bleu.py`	`translation_fidelity_bleu.json`
§5.3 NLLB†	`probe_nllb_translation.py`	`nllb_probe_*.json`
§5.3 few-shot†	`probe_fewshot_translation.py`	`fewshot_probe_*.json`

Table 6: Each reported table, the command that regenerates it from the released artifacts, and the artifact it reads. Every row but the two marked † is computed from the saved generations and needs no model: those two run a model, NLLB-200 and gemma-7b respectively, and we release their outputs so the reported numbers can be checked without doing so.

dropping 7 duplicate (English, Tigrinya) pairs and 1 row with no English side.

From 5,775 to 5,701. The two tagging tasks were run against a gold file in which each multi-sense entry is represented by its first sense: (*a little*, ንእሽቶይ, ቁሩብ, ውሕድ) was evaluated as (*a little*, ንእሽቶይ). That intermediate file was not originally released, so the task scripts as published would evaluate 5,775 items rather than the 5,701 behind Sections 5.1 and 5.2. We reconstruct it from the released gold and verify the result row-for-row against the saved model outputs: 5,701 items, no row missing, none extra, and no label mismatch. The reconstructed file is released alongside the gold.